

GYULA (*Jules*) PLANKY

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TECHNICAL SKILLS

Programming Languages: Go, Python, JavaScript, SQL, Bash; Java Perl

Frameworks: FastAPI, Flask, Django, React, Node.js, Express, gRPC

ML / AI: PyTorch, TensorFlow, Hugging Face, scikit-learn, Pandas, NumPy, MLflow

Databases: MySQL, PostgreSQL, MongoDB, DynamoDB, Redis

Infrastructure / Tools: Docker, Kubernetes, Terraform, AWS, GCP, Airflow, Jenkins, GitHub Actions,

EXPERIENCE

- Morgan Stanley Capital International**

- Fixed Income Reference Data Analyst*

2022 – 2023

- Maintained production Perl/Python pipelines on Linux processing **3M+ instruments daily** with **99.999% uptime**, serving **7,000+ clients** across **100+ countries**.
- Improved pipeline performance by **20%** through SQL query optimization and connection pooling, reducing processing time from **45 to 30 minutes**.
- Led Python migration project with **85% test coverage**, writing Bash deployment automation and implementing monitoring system that caught **95%** of data quality issues before client impact.
- Built ETL features integrating **50+ new instruments**, cutting manual processing time by **50%** through SQL-based automation.

- Morgan Stanley**

- Treasury Analyst*

2021 – 2022

- Built Power BI dashboards with **Python backend** automating capital and liquidity monitoring across **20+ legal entities**, reducing manual reporting time by **40%** and supporting **\$50B+ daily** regulatory compliance workflows.
- Replaced **10+ Excel/VBA macros** with Python scripts using Pandas, cutting daily report generation from **2 hours to 15 minutes**.

- Vasutville / Falkenburg**

- M&A Intern / Controller Intern*

2019 – 2020

- Conducted due diligence on **5+ M&A projects** ranging **\$1-20M** each; built DCF models and sensitivity analyses in Excel, streamlining IFRS-compliant reporting.

TECHNICAL PROJECTS

- **NEUDFS – Distributed File System** – Built a cloud-native distributed file system in **Go** with **gRPC/Protobuf**, serving a university environment with RBAC (students, professors, TAs). Stored file data in **S3** and metadata/permissions in **DynamoDB**; deployed on **AWS ECS** behind an NLB with **Terraform**-managed infrastructure. Achieved **sub-1ms p95 latency drift** under sustained concurrent load in soak tests.
- **Search Ranking w/ Transformers** – Designed machine learning search relevance system for e-commerce platform with **10k+ products**. Built embedding-based retrieval using **BERT** and custom re-ranking models (LambdaMART, neural networks), improving search result quality by **20%**.
- **Bluebikes Demand Prediction System** – Production ML pipeline forecasting actual bike demand in **Boston, MA**. Trained ensemble model (**R²=0.87**, **12 bikes/hour error**), deployed FastAPI service with **sub-200ms latency** on AWS ECS. Implemented automated retraining with Airflow and drift detection with Evidently.
- **Self-Hosted NAS System** – Deployed Raspberry Pi 5 NAS serving **1TB storage** across **10+ devices** with NFS/Samba shares. Configured Tailscale mesh VPN for secure access achieving **20MB/s throughput**.

EDUCATION

- Northeastern University, Khoury College of Computer Sciences**

Boston, MA

- Master of Science (MS) in Computer Science*

Sept 2024 – May 2026

Coursework: Algorithms, OOP, Distributed Systems, Cloud Computing, Deep Learning, ML Ops

- ESCP Europe**

Paris, Madrid, Berlin

- Bachelor of Science (BS) in Business Management*

2017 – 2020

ADDITIONAL

Languages: English ■■■ Hungarian ■■■ Spanish ■■■ French ■■■ German ■■■

Interests: Hardware prototyping (Arduino, RaspberryPi), wood/metal working, sailing, surfing, climbing